№ T12

$\alpha = 0{,}2$; $n = 2$

$H_0: \ \xi \sim \frac{1}{3}\{4\} + \frac{1}{6}\{3\} + \frac{1}{4}\{1, 2\}$

$H_1: \ \xi \sim \frac{1}{4}\{1, 2, 3, 4\}$

„3" выпало m раз, „4" – s раз;

$\Rightarrow$ „1" или „2" – $n - m - s$ раз

$$l(\vec{x}_2) = \frac{L_1}{L_0} = \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{3}\right)^s \left(\frac{1}{6}\right)^m \left(\frac{1}{4}\right)^{2-s+m}} = \frac{\left(\frac{1}{4}\right)^{s+m}}{\left(\frac{1}{3}\right)^s \left(\frac{1}{6}\right)^m}$$

m, s	$0,0$	$0,1$	$0,2$	$1,0$	$1,1$	$2,0$
l	1	$3/4$	$9/16$	$3/2$	$9/8$	$9/4$
$H_0 \to P$	$1/4$	$1/3$	$1/9$	$1/6$	$1/9$	$1/36$
$H_1 \to P$	$1/4$	$1/4$	$1/16$	$\frac{1}{4}$	$1/8$	$1/16$

$P(l \geq C \mid H_0) \leq \alpha = 0,2$

① : $(2,0)$ $\quad P = \frac{1}{36} \leq 0,2$ ✓

② : $(1,0)$ $\quad P = \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \leq 0,2$ ✓

③ : $(1,1)$ $\quad P = \frac{1}{36} + \frac{1}{6} + \frac{1}{9} = \frac{11}{36} > 0,2$ ✗

$\Rightarrow$ крит. обл-ть: „выпала хотя бы одна 3."

$W = P(l \geq C \mid H_1) = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}$